Module Assignment & Exam Outlines

Module code and Name	**ELEC70126 - Internet of Things and Applications**
Module Leader	**Hamed Haddadi**
Assessment name & weighting	**Coursework, 100%**
Assessment date	**2pm Weds 25/03/2026 (submit once only) (optional half-page work-plan to be submitted by 11/02/2026 for feedback, this piece is not part of the final marks)**

Summary description and objectives for the assessment

Part 1: Sensing (50%)

对于至少持续一周的来自至少 2 个不同性质的时间序列数据源（例如，来自传感器的环境数据、移动传感器数据、社交媒体数据、智慧城市 API 数据等）的数据，建立一个平台以实现自动执行以下操作：

For at least a week's worth of continuous data from at least 2 time-series data sources of different nature (e.g., ambient data from sensors, mobile sensors, social media, smart city APIs, etc), set up a platform to automatically:

1. Acquire data from a custom prototype sensor or a public data source
2. Establish a network communications and/or data storage to save the data
3. Justify the design choices, and discuss any data cleaning and/or filtering

1. 从定制的原型传感器或公共数据源获取数据
2. 建立网络通信和/或数据存储系统以保存数据
3. 说明设计选择的合理性，并讨论任何数据清理和/或筛选的操作。

Part 2: Analytics and interface (50%)

第二部分：分析与界面（50%）
利用第一部分所获取的数据，设计并实现以下内容的原型：

Use the data from part 1 and design and implement a prototype for:

1. A web App/phone app/other user interface/interaction form (wearable, smarthome, etc) to present the data and provide user engagement opportunities,
2. Perform Basic time series data analysis to provide added insight into the data,
3. Present and discuss opportunities Innovation, creativity, enterprise, scalability & complexity analysis within your ecosystem

1. 一种网络应用程序/手机应用程序/其他用户界面/交互形式（可穿戴设备、智能家居等），来展示数据并提供用户参与的机会。

2. 进行基本的时间序列数据分析，以进一步深化对数据的理解。

3. 当前并探讨以下机会：在您的生态系统内，涉及创新、创造力、创业精神、可扩展性及复杂性分析等方面的内容。

Each individual must submit their own assignment independently.

每位成员都需独立完成自己的任务。

Submission requirements (for assignments only)

- Submissions: 1 report to be submitted per student, combined PDF of up to 10 pages, one for part 1 and one for part 2 (up to 5 pages each), using the template on Blackboard

- Code & Data: link to accessible code (ideally Github, or publicly accessible Dropbox/OneDrive/etc folder), link to data repository

Learning outcomes (from module descriptor)	Assessment criteria

Having an understanding of the basic principles of Sensing, IoT, and digital signal processing	Correctness, appropriateness and depth of the technical acoustics and digital signal processing features within the developed work.
Haveingan understanding of the physical and social sensing systems and of the main principles of Internet of Things	Appropriateness and depth of the developed work in relation with the broader Internet of Things landscape.
Be able to integrate different sensor data software and hardware components, designing and creating actuations, analytics domains, products and/or complex interactions/experiences	Complexity and correctness of the integration between hardware and software components within the developed work. This criterion will be assessed considering also the use of software components in the design stage to predict and emulate the final product. If the work includes only software components, this criterion will be assessed considering also the graphic user interface of the developed work.
Have an experience of applying such knowledge to appropriate case studies	Appropriateness and depth of the solution found to solve the specific assigned tasks. Appropriateness of the planning and timeliness of the delivery. Appropriateness and depth of the reporting

Assessment criteria	Pass (D grade)	Excellent (A grade)	Weight (%)
Correctness, appropriateness and depth of the technical set up and digital signal processing features within the developed work.	Generally correct and appropriate features, but without technical depth.	The work shows features that are appropriate and correct considering the specific task, and that show a clear technical depth in terms of analysis and development.	30%
Appropriateness and depth of the developed work in relation with signal cross-correlation.	Basic but correct considerations of correlations of physical and social signals, without in-depth analysis.	Appropriate and extensive considerations of the time-series correlations and/or perception within the developed work	10%
Complexity and correctness of the integration between hardware and software components within the developed work.	Basic but appropriate integration between hardware and software components, aimed only at linking the two, and without	Correct and extensive integration of the hardware and software components.	10%

	considering the full potential of the interaction.		
If the work includes only software components, this criterion will be assessed considering also the graphic user interface of the developed work.	Basic but appropriate graphical user interface.	Appropriate graphical user interface which shows extensive work in the design, ease of use and access of the various component.	
Appropriateness and depth of the solution found to solve the specific scenario.	Basic but appropriate solution, correctly implemented.	Appropriate and correct solution, which demonstrates extensive depth in terms of analysis and technical features.	20%
Appropriateness of the planning and timeliness of the delivery.	Generally appropriate planning, with an overall timely delivery of the various components, without in-depth considerations about the various processes.	Appropriate and extensive planning, with timely delivery of the various components and in-depth considerations of the various stages of the design, development and delivery.	10%
Appropriateness and depth of the reporting	Appropriate report, containing required multimedia material, sufficiently well organised but without in-depth information about the various stages of the work. No in-depth considerations on the overall installation.	Appropriate and extensive report, containing all the required multimedia material, all appropriately organised and presented, and with in-depth information about the various stages of the design, planning, development and delivery of the work. In-depth considerations on the overall installation.	20%

Appendix: Guidance questions

For the various section of the report, think about the following questions:

• Executive summary:
o Have you outlined the background to the problem?
o Have you clarified the purpose of the report?
o Have you provided brief details of the approach you followed? o Have you summarised the important results and findings?
o Have you stated the major conclusion from your work?

• Design and implementation.
o Have you identified the problem you are addressing?
o Have you provided an overview of how you are addressing this problem?
o Have you provided detailed steps of what you did in a logical order?
o Have you described the methods you used to solve the problem and their purpose? o Have you identified the parameters for designing your system?
o Have you identified the metrics for evaluating the performance of your system?
o Have you used illustrations to give the reader a visual interpretation of your system?

• Results.
o Every result included must have a method set out in the design methodology.

o Likewise, every method should have some results.
o Have you used figures and tables to illustrate your points?
o Have you critically evaluated the quality and reliability of your results?
o Are your observations supported by evidence?
o Is the analysis and discussion relevant to the project and its context?
o Have you explained how well (or not) you have met the project's objectives?

• Trade-offs.
o Have you considered and quantified the cost (you will incur) versus the gain (you will

obtain) when you modify different parameters in your system? For example: ▪ Local vs. Cloud processing.
▪ Range vs. Power consumption (battery life).
▪ Throughput vs. Power consumption.

▪ Latency / Delay vs. Accuracy.
▪ Performance vs. Security.
▪ Generic vs. Bespoke hardware. ▪ Price vs. Functionality.

▪ Price vs. Usability.
▪ Price vs. Robustness / Reliability.
▪ Functionality vs. Usability.
▪ Functionality vs. Rapid development (time to market). ▪ Functionality vs. Backward compatibility.
▪ Flexibility vs. Accuracy vs. Interpretability.